



Customer Purchase Analysis & Insights

From raw data to actionable business intelligence using Python, PostgreSQL, and Power BI

Project Overview



Business Problem

Convert retail transaction data into meaningful insights for improved sales and customer experience



Key Objectives

Analyze purchase behavior, identify customer segments, determine profitable products, understand retention factors



Technology Stack

Python (Pandas, NumPy), Google Colab, PostgreSQL (Supabase), PgAdmin, Power BI

Dataset

3,900 customer transactions with 18 variables including demographics, purchase history, product details, and ratings

Process

Data cleaning → Feature engineering → SQL analysis → Power BI visualization

Data Preprocessing & Cleaning

- Data Loading

CSV imported using Pandas, initial inspection revealed
3,900 rows × 18 columns

- Missing Values

37 null ratings imputed using median by category,
preserving dataset size

- Standardization

Column names converted to lowercase with underscores
for consistency

- Feature Engineering

Age groups created, purchase frequency converted to
days, redundant columns removed

Power BI Dashboard

Connection Process

- Supabase PostgreSQL → PgAdmin verification
- SSL configuration for secure connection
- Power BI import via Get Data option
- Live connection to PostgreSQL database

Dashboard Components

5 KPI cards, 4 slicers (Gender, Subscription, Shipping, Discount), interactive visualizations with drill-down capabilities



Data Analysis using SQL

Exploring key business insights through structured SQL queries and data analysis.

1. Top 3 Most Purchased Products Within Each Category

rank	category	item_purchased	purchase_count
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
2	Accessories	Belt	161
3	Accessories	Scarf	157
1	Clothing	Blouse	171
1	Clothing	Pants	171
2	Clothing	Shirt	169
3	Clothing	Dress	166
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

2. Top 5 Products with the Highest Average Review Rating.

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

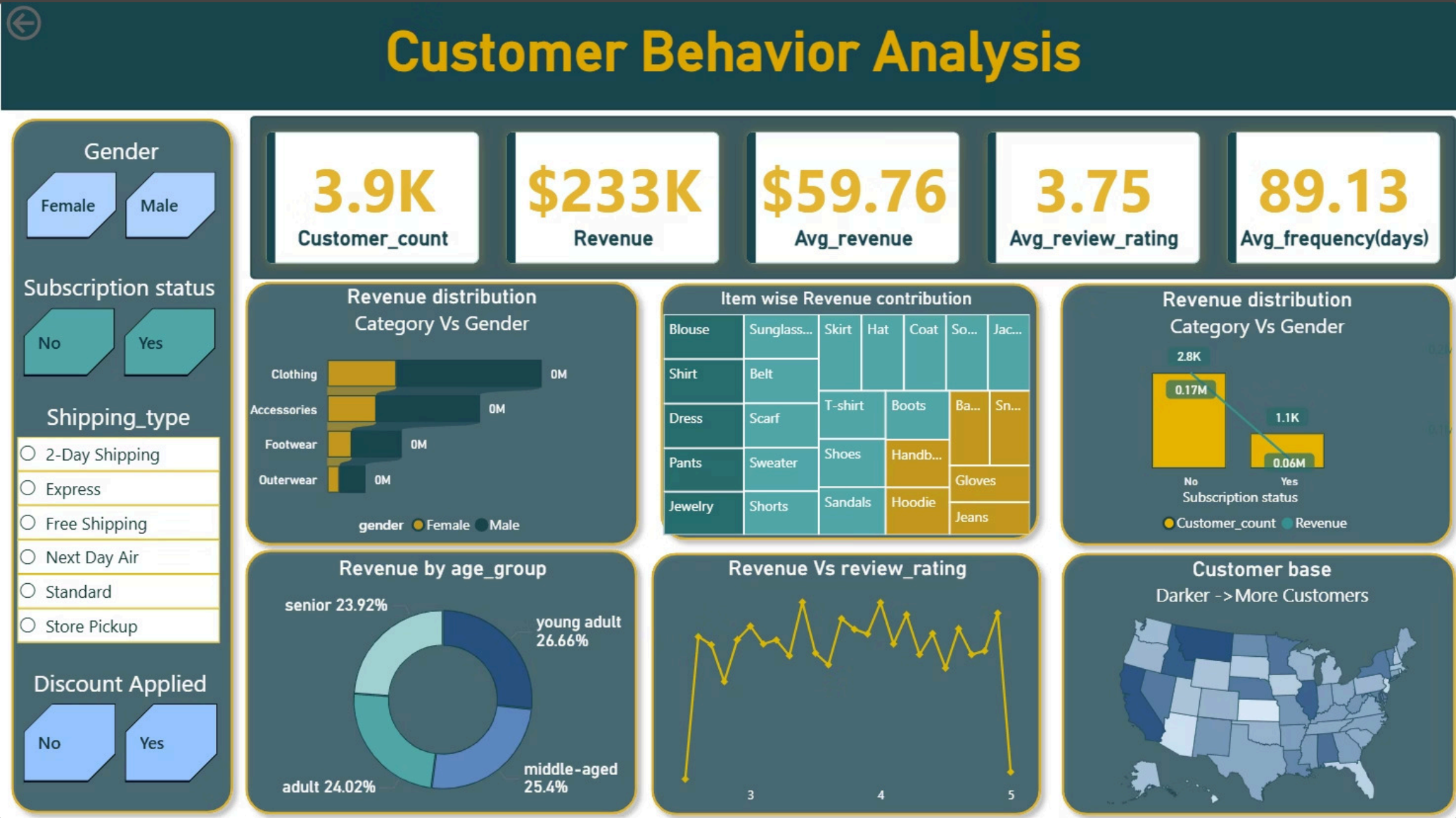
4. Customer Segmentation: New, Returning, and Loyal

segment	count
Loyal	3116
New	83
Returning	701

3. Do Subscribed Customers Spend More?

subscription_status	customer_count	avg_purchase_amount	total_revenue
No	2847	59.87	170436.00
Yes	1053	59.49	62645.00

Dashboard



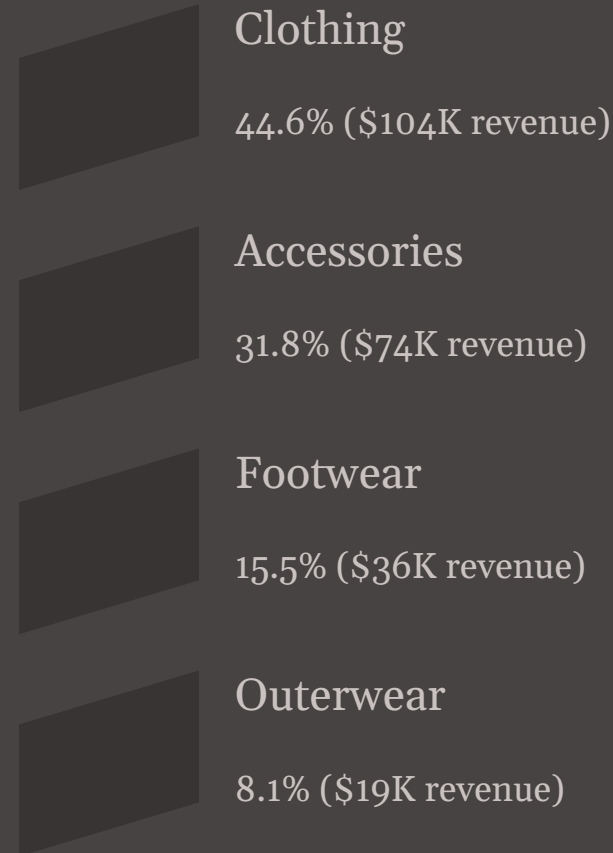
Key Findings: Revenue & Categories

Slicers Used

- Gender
- Subscription Status
- Shipping Type
- Discount Applied

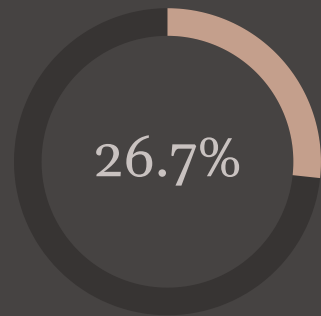
Performance Metrics:

- **Total Revenue:** \$233K from 3,900 transactions
- **Avg Transaction:** \$59.76 per purchase
- **Avg Rating:** 3.75 out of 5.0
- **Customer Count:** 3.9K
- **Avg Frequency:** 89.13 days



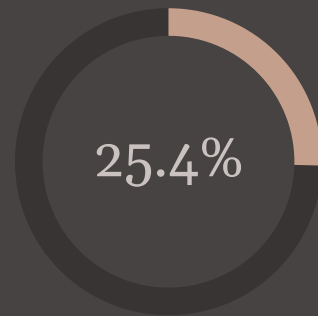
Insight: Clothing + Accessories = 76% of total revenue. Male customers contribute higher across all categories.

Customer Segmentation Insights



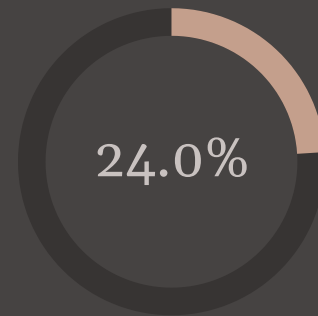
Young Adult

Revenue contribution



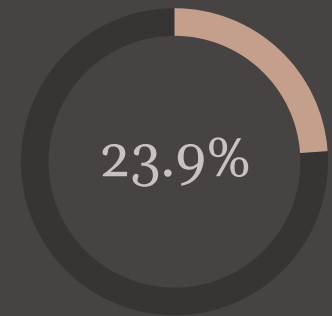
Middle-Aged

Revenue contribution



Adult

Revenue contribution



Senior

Revenue contribution

Purchase Frequency

Average: **89 days** between purchases

Customer Base

Top states: Montana, California, Idaho, Illinois, Alabama

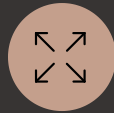
Strategic Recommendations & Operational Actions

Strategic Recommendations



Top States Priority

Montana, California, Idaho, Illinois, Alabama — targeted digital ads and seasonal campaigns



Expand Low-Density

Introductory discounts and free shipping in untapped regions



Clothing Focus

Prioritize inventory and promotions for top revenue categories



Male Engagement

Dedicated strategy with personalized recommendations

Operational Actions



Improve Ratings

Implement proactive feedback loops and quality assurance checks



Reduce Gap

Streamline logistics to close the purchase frequency time gap



Bundle Products

Create thematic product bundles to increase average order value



Strategic Conclusion

This project reveals significant opportunities to drive growth by focusing on high-performing segments and addressing identified operational gaps. By optimizing digital presence in key states and refining our product mix, we can effectively capture untapped market share.

Key Takeaways

- **Revenue Drivers:** Prioritizing top categories like clothing and engaging male demographics will yield the highest returns.
- **Regional Growth:** Targeted expansion into low-density areas with localized campaigns is essential for scale.
- **Operational Excellence:** Closing purchase frequency gaps through proactive engagement and bundle strategies will increase long-term customer lifetime value.

Moving forward, our immediate focus shifts to implementing these targeted campaigns and streamlining logistics to ensure sustainable, data-driven growth.